

Literature Status: One-Factor Projective Tensor Octahedrality

Automatic Math Discovery Run `fa_banach_001`

June 28, 2026

Status

This is a **literature_already_answered** packet. It records that a later paper explicitly answers one of the open questions extracted from arXiv:1609.02062. It is not a new proof packet.

Source Question

The source paper is:

Juhan Langemets, Vegard Lima, and Abraham Rueda Zoca, *Octahedral norms in tensor products of Banach spaces*, arXiv:1609.02062.

Near the end of the paper, after proving a partial positive result for projective tensor products with a non-reflexive L -embedded factor, the authors write that for general Banach spaces X and Y it remains open whether

$$X \widehat{\otimes}_{\pi} Y$$

has an octahedral norm whenever X and/or Y has an octahedral norm. The source also describes the partial result as a partial positive answer to [1, Question 4.4], where the one-factor projective preservation problem is asked.

Supporting Answer

The supporting paper is:

Abraham Rueda Zoca, *Stability results of octahedrality in tensor product spaces*, arXiv:1708.07300.

The abstract of the supporting paper states that there exists a finite dimensional complex Banach space X such that

$$L_1^{\mathbb{C}}([0, 1]) \widehat{\otimes}_{\varepsilon} X$$

fails the strong diameter two property and

$$L_{\infty}^{\mathbb{C}}([0, 1]) \widehat{\otimes}_{\pi} X^*$$

fails to have an octahedral norm. It further states that this proves that octahedrality is not automatically inherited from one factor by taking projective tensor product, answering [1, Question 4.4].

The paper proves this through Theorem 2.1 and Theorem 2.2. In particular, Theorem 2.2 gives a two-dimensional complex Banach space Y such that

$$L_{\infty}^{\mathbb{C}}([0, 1]) \widehat{\otimes}_{\pi} Y$$

does not have an octahedral norm. The following remark identifies why this is a counterexample to the one-factor problem: $L_{\infty}^{\mathbb{C}}([0, 1])$ has the Daugavet property, hence has an octahedral norm, but the projective tensor product above is not octahedral.

Identification

The final open question in arXiv:1609.02062 asks whether octahedrality of one factor forces octahedrality of the projective tensor product in general. The supporting paper constructs a projective tensor product

$$L_{\infty}^{\mathbb{C}}([0, 1]) \widehat{\otimes}_{\pi} Y$$

where the first factor is octahedral but the tensor product is not. This is a negative answer to the same one-factor projective tensor product question.

The bibliographic identification is explicit: arXiv:1708.07300 cites [1] as Question 4.4 and cites the arXiv:1609.02062 source paper as [2]. Thus this packet is best classified as `literature_already_answered`, not as a new counterexample.

Scope Limitations

This packet clears only the projective tensor product octahedrality question. It does not claim a new proof of the source paper’s partial positive theorem, and it does not address unrelated diameter-two or injective tensor product stability questions except insofar as they are used in the supporting paper’s duality argument.

Search Evidence

Local cheap indexes for arXiv:1609.02062, the title, octahedral norms, projective tensor products, and Question 4.4 had no prior exact packet at reservation time. A targeted literature search found arXiv:1708.07300, whose abstract and post-Theorem 2.2 remark explicitly state the negative answer to Question 4.4.

References

- [1] J. Langemets, V. Lima, and A. Rueda Zoca, *Almost square and octahedral norms in tensor products of Banach spaces*, Rev. R. Acad. Cienc. Exactas Fis. Nat. Ser. A Mat. **111** (2017), 841–853.
- [2] J. Langemets, V. Lima, and A. Rueda Zoca, *Octahedral norms in tensor products of Banach spaces*, Q. J. Math. **68** (2017), no. 4, 1247–1260; arXiv:1609.02062.
- [3] A. Rueda Zoca, *Stability results of octahedrality in tensor product spaces*, arXiv:1708.07300.